(a) Entropy-coded transmission and recovery

Transmitter (Tx)

Frozen policy lookup
Input: (N, SNR)
Target (m*, K*); MCS (M, r)
Calibrated offline; queried per image

Target endpoint / fallback chain

Frozen MCS (M, r)



Image x
(256 × 256 × 3)

Frozen VAR tokenizer
- **Visual encoder** $E(\cdot)$
- **Multi-scale residual VQ** (shared codebook)
- Frozen

Multi-scale tokens $\{r_1, \ldots, r_{10}\}$

$s_{10} = 16$

Prefix selection
All scales 1, …, m + first K in scale m + 1 (raster order)

Actual-length check
$L_{\text{term}} + 13 + 16 \le k_{\text{LDPC}}$
Else: step down the frozen fallback chain

Actual endpoint (m, K)

Digital transmitter

Image payload (single LDPC codeword)
VAR-conditioned arithmetic encoder
MCS: (M, r)
Length 13 bits | Terminated AC stream | Zero padding | CRC16 → LDPC → QAM

Protected public header · 68 symbols
Actual profile ID (m, K, M, r, …) → CRC → Conv. coding + QPSK

Frame assembly

AWGN channel (known SNR)
$y = s + n$

Receiver (Rx)

Data — Control

Same frozen VAR prior at Tx and Rx
CDFs reconstructed locally from token history

Digital receiver
- Header decode → actual profile
- QAM demap → LDPC decode
- CRC / length check
- Arithmetic source decoding

Header / CRC / length / AC failure → **fixed gray output**

Decoded token prefix

VAR completion (frozen)

Codebook lookup rescale + residual sum

Image decoder $D_c(\cdot)$ (frozen)



Reconstruction $\hat{x}$
(256 × 256 × 3)

Thumbnails are schematic

Entropy source-bit utilization · 10 dB · 500-source means

AC budget: 2520 = 2549 − 13 − 16 bits

WHOLE (whole scales)	**77.70% AC source bits**	22.30% padding
PARTIAL (within-scale)	**95.21% AC source bits**	4.79% padding

Same physical frame for both
68 | **956 coded-body symbols**
68 + 956 = 1024 channel symbols

(b) Boundary-scale recovery: entropy-coded example, 10 dB, m = 7, K = 75

Decoded scales 1–7
155 token indices

s_1 1 × 1 | s_2 2 × 2 | … | s_7 8 × 8

VAR predicts all scale-8 tokens
10 × 10 = 100
argmax

75 decoded scale-8 tokens

Overwrite first 75 positions (raster order)

Scale 8 (merged)
75 decoded + 25 predicted = 100 tokens (10 × 10)

Advance with merged scale-8 state

Scale 9 (predicted)
13 × 13 = 169 (all predicted)

Scale 10 (predicted)
16 × 16 = 256 (all predicted)

Decoded token
Predicted token